Reflection on Group Assignment 2IRR00

Amber Slob
Eindhoven University of Technology, The Netherlands
2259656, Group 24

1 Project Description and Scope

Our project aimed to build an easy-to-use audio-book player. The main goal was to make the application accessible. Therefore, we minimized the number of buttons and unnecessary functionalities and implemented the possibility to change the color scheme. The program must allow users to select, play and pause audio-books and upload or remove audio-books. It should also be possible to adapt the sound-playing speed and volume of the audio-books, according to preference, and the program should remember the progress. Furthermore, users could be able to set sleep-timers, turning off the audio-books after a certain amount of time and adjust the font size and color of the app. We have managed to implement all of our main functionalities, mentioned above.

2 Project Reflection

2.1 Process and Collaboration

Our group consisted of six people. Initially, our communication went well. We held meetings and assigned a chair and a note taker and managed to create a clear overview of our project and functionalities. Unfortunately, contact did not continue to go smoothly during the rest of our project. Only two to three people usually showed up to instructions which moved our discussion to text and thus delayed responses and led to an increase in miscommunication.

Regarding our task distribution of the implementation, we also encountered some issues. We immediately defined task distributions. However, we did not set clear internal deadlines and failed to properly coordinate dependencies between tasks. Eventually, we assigned two people to manage reviewing pull requests, created branches and everyone could work on an aspect of the audio-book player and its functionalities in parallel. Furthermore, we set internal deadlines to ensure there was enough time to resolve problems.

2.2 Testing and Quality Assurance

During this project, I realized the usefulness of testing and quality assurance. When defining tests, it helps me see how the code should function. I have found that forcing myself to stop programming and consider what cases the code should be able to handle and how, is a very effective debugging strategy. Furthermore, I have realized the importance of checking input and throwing exceptions.

2.3 Design Principles and Patterns

We have implemented four design patterns; Singleton, Strategy, Template Method and Model-View-Controller. The Singleton design principle was useful to work with. I used Settings to make the Audiobook selected in AudiobookOverview available to the Runnable episodeView. This worked as all classes shared the same instance of Settings. I have not worked on the interfaces implementing the Strategy pattern or the Template Method. Finally, I was not informed that we wished to follow MVC and my GUI was not initially implemented as an FXML view. I did, however, use CSS for styling. Design principles and patterns are concepts I have only recently learned, so it is interesting to see them immediately in practice.

3 Self-Reflection

My main contribution were making the Use-case diagram and activity diagram, with advice from Vincent, building the audio-book overview and I have connected pages together through the buttons. Furthermore I held the presentations, worked on documentation and I went to the planned meetings and all instruction. I managed to complete all tasks assigned to me well. One aspect I could improve on is my communication, as I have had conflicts regarding this. In one case, my teammate asked me to implement a functionality the same day, while I was not available this day. However, I failed to properly communicate this. Furthermore, I have once missed an internal deadline by 2 hours, due to miscommunication.

4 Conclusion and Learnings

During this project, I have learned a variety of new skills. Organizing meeting, creating diagrams, integrating design principles into software and new Java functionalities, such as Runnables. I believe this course has already improved my coding abilities significantly. Furthermore, I realized the value of design principles and patterns, as well as testing and quality assurance. Finally, learning how to properly structure meetings has been an important lesson.